ABSTRACT OF THE DISCLOSURE

An adaptive troposcatter communication system for transmitting high-radix, parametric data over variable beyond-line-of-sight (BLOS) radio channels. The system comprises adaptive transceivers that continuously monitor the troposcatter channel's conditions. Based on this real-time analysis, the transceivers dynamically adjust a symbol-to-state mapping that defines a set of N discrete, distinguishable RF states (where N greater than 2) used to encode abstract symbols. The system can vary the radix (N) of the transmission, reducing it to enhance reliability during poor channel conditions and increasing it to maximize throughput when conditions are favorable. Data is transported as unevaluated parametric expressions, ensuring end-to-end data fidelity is maintained across the inherently unstable troposcatter link.